

Wildlife and Species Richness of Canada

Canada's legacy to future generations will be the protection of Canada's native wildlife and its species richness. What does this mean? A legacy is something, material or immaterial, that is preserved by one generation for another. The Canadian government has established COSEWIC (The Committee on the Status of Endangered Wildlife in Canada) as a legal entity to assess the level of risk of extinction for wildlife species. It is based on the best available, scientific, Aboriginal traditional, and community knowledge on the status of the species of birds, mammals, amphibians, plants, fish and others that are native to Canadian land and waters. COSEWIC is an independent committee made up of wildlife experts from several universities and museums, provinces and territories, three national conservation organizations (Canadian Nature Federation, Canadian Wildlife Federation, World Wildlife Fund Canada) and four federal agencies (Canadian Wildlife Service, Parks Canada, Fisheries and Oceans Canada, Canadian Museum of Nature). It is their responsibility to ensure that native species and their natural environment are protected so that they can be sustained and produce future generations. The task of this committee is to determine whether any particular species is at risk and if so, to what extent. Whenever any of these species become endangered, threatened or vulnerable they are placed on a COSEWIC list. In 1999 there were 339 species on the COSEWIC list. This list is constantly changing; new species are added, some species are moved to different levels of risk when their situation either improves or deteriorates, some are removed all together if they are no longer at risk. They are assessed at seven levels of risk from greatest to lowest:

- *extinct* - a species that no longer exists
- *extirpated* - a species that no longer exists in the wild in Canada, but occurs elsewhere, (for example, in captivity or in the wild in the United States).
- *endangered* - a species facing imminent extirpation or extinction
- *threatened* - a species likely to become endangered if limiting factors are not reversed
- *special concern* (formerly called vulnerable) - a species of special concern because of characteristics that make it particularly sensitive to human activities or natural events
- *species not at risk* - a species that is not in danger of extinction
- *data deficient* - insufficient data on population size and distribution, on geographic range or regarding sensitivity to human activities and threats.

There are four main questions asked to determine whether a species is at risk.

1. Is it a taxonomically valid species, subspecies/ variety or population according to taxonomical experts?
2. Is it a species native to Canada (either naturally occurring or having persisted here for at least 50 years)?
3. Is it a species of regular occurrence in Canada, not an accidental visitor?
4. If it is a non-resident or migratory species, does it require Canadian lands or waters for a key part of its life cycle?

The importance of protecting our wildlife is emphasized by the number of species appearing on the COSEWIC list. Canada is fortunate to be rich in flora and fauna. There are more than 150

tree species in Canada and the most recent list of birds found in Canada has 417 different species on it. There are 186 native mammals across the 15 terrestrial and 5 marine ecozones, and 88 reptiles and amphibians.

The occurrence of any one of these species within an ecozone is determined by its range. A *range* is the geographical limits of the area of land or water in which the species has been found and which supports its habitat requirements. Some vertebrate species have very small ranges and remain there year-round. An example might be a particular species of fish that is limited to a few lakes. Others are migratory and have different ranges at specific times of the year. For example, geese have a breeding range in the northern ecozones, a summer range across much of Canada and a winter range mostly south of the border in a climate warmer than that found in Canada. Other mammals, such as big horn sheep, occupy relatively small areas and have summer ranges at higher altitudes and move down into sheltered valleys in the winter where food is more abundant.

The number of species found within any ecozone is largely dependent on the climate and the availability of food sources. This not only applies to the vertebrate species but to plant species as well. In the southern ecozones where the soil is rich, plant life flourishes and consequently so does herbivores (plant-eaters: e.g. deer, bear, rabbit), insectivores (insect-eaters: e.g. birds, skunks) and grainivores (grain-eaters: birds, rodents). In the progression of the food chain, the carnivores are also abundant in those regions. However, in the farthest northern ecozones where the surface is mostly bare rock, very few species of plant or animal can survive. Most mammals, which can survive in these regions, depend on fish or smaller fish-eating mammals. It is to these northern regions that the migratory species, mostly birds, go to breed their young in spring and raise them during the short summer. Most of Canada's wetlands are found in a broad band around Hudson Bay and these areas produce abundant insect populations on which the birds feed. The table below shows the number of terrestrial species found within each of the ecozones.

Table 1: Terrestrial Species within each of the Ecozones

Ecozone names	Bird	Mammal	Reptile/ Amphibian	Tree	Total Species	Total minus COSEWIC
Arctic Cordillera	69	29	1	6	105	96
Northern Arctic	72	23	0	0	95	83
Southern Arctic	141	44	1	12	198	184
Taiga Plains	212	57	7	20	296	284
Taiga Shield	193	53	9	23	278	264
Boreal Shield	267	71	43	80	461	398
Atlantic Maritime	195	54	38	64	351	309
Mixedwood Plains	238	58	51	106	453	325
Boreal Plains	278	81	21	49	429	404
Prairies	280	84	26	44	434	388
Taiga Cordillera	174	47	1	13	235	229
Boreal Cordillera	194	52	6	28	280	270
Pacific Maritime	250	85	31	47	413	358
Montane Cordillera	265	92	33	52	442	400
Hudson Plains	185	50	9	26	270	262

Species biodiversity refers to the variety and number of species found within an ecozone. It is affected by a variety of factors. We must consider the number of species there are in an ecozone to begin with. Then we will consider the number of species within that ecozone that are on the COSEWIC list. Other factors, which might affect the survival of the species within an ecozone, are the population density and the density of roads. In reality, there are many more factors involved and to a much more complex level, but for our purpose of learning how modeling works, we will keep it simple. Modeling uses mathematical equations to rank the importance of various factors. In our case, total number of species, number of COSEWIC species, population density and road density will be considered as to the effect they have on the diversity of the species.

In order to protect our natural environment, the Canadian government has legislated several laws to protect native Canadian species from ending as endangered species. Below, you will find part of the legislation that has gone before Parliament. These regulations protect not only the species as individual organisms but also their natural habitat. They ensure that each protected species is given the best chance at survival.

Part of Bill C-5

MEASURES TO PROTECT LISTED WIDLIFE SPECIES

General Prohibitions

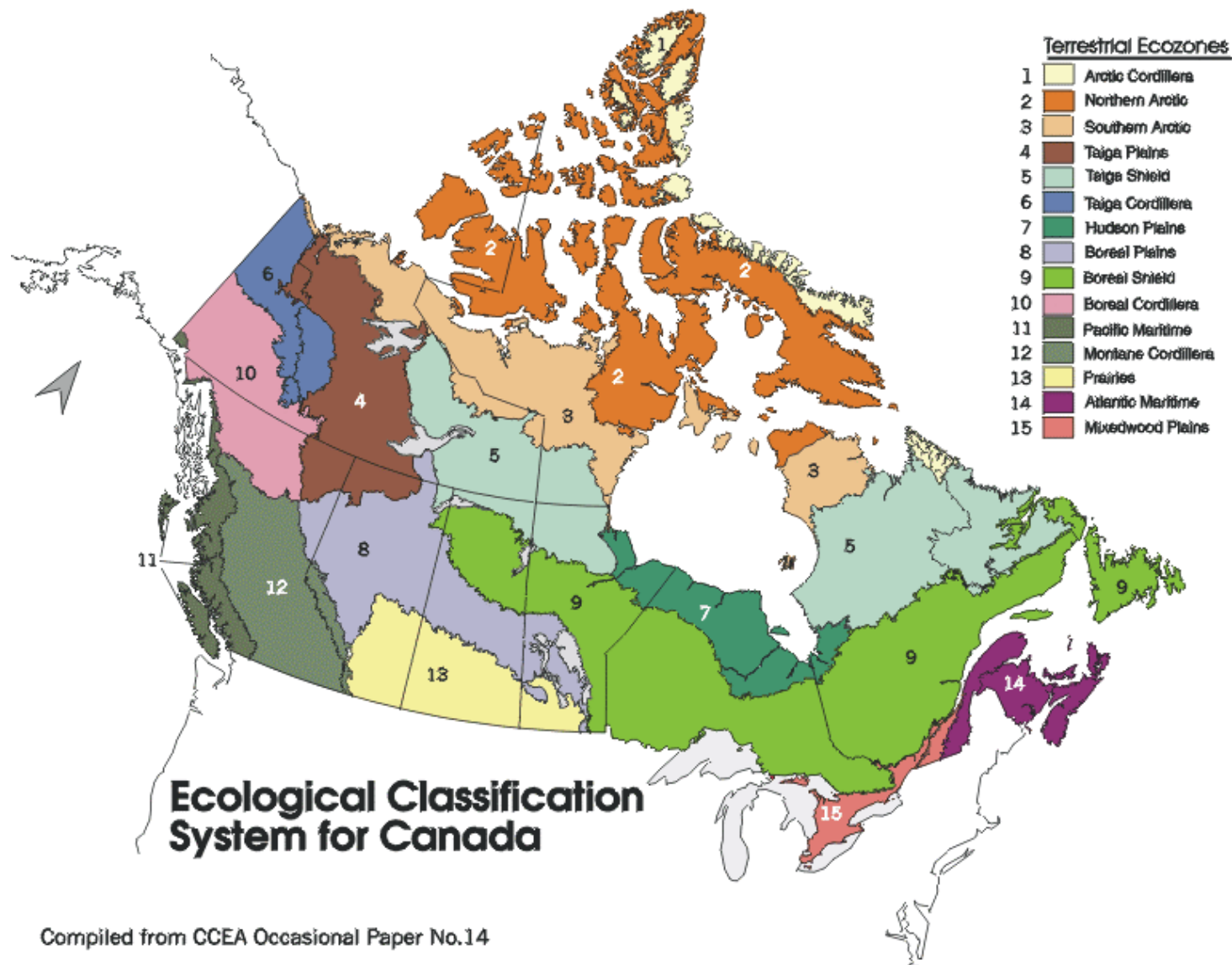
Killing, harming, etc., listed wildlife species	32. (1) No person shall kill, harm, harass, capture or take an individual of a wildlife species that is listed as an extirpated species, an endangered species or a threatened species.
Possession, collection, etc.	(2) No person shall possess, collect, buy, sell or trade an individual of a wildlife species that is listed as an extirpated species, an endangered species or a threatened species, or any part or derivative of such an individual.
Deeming	(3) For the purposes of subsection (2), any animal, plant or thing that is represented to be an individual, or a part or derivative of an individual, of a wildlife species that is listed as an extirpated species, an endangered species or a threatened species is deemed, in the absence of evidence to the contrary, to be such an individual or a part or derivative of such an individual.
Damage or destruction of residence	33. No person shall damage or destroy the residence of one or more individuals of a wildlife species that is listed as an endangered species or a threatened species, or that is listed as an extirpated species if a recovery strategy has recommended the reintroduction of the species into the wild in Canada.
Application - certain species in provinces	34. (1) With respect to individuals of a listed wildlife species that is not an aquatic species or a species of birds that are migratory birds protected by the <i>Migratory Birds Convention Act, 1994</i> , sections 32 and 33 do not apply in lands in a province that are not federal lands unless an order is made under subsection (2) to provide that they apply.
Order	(2) The Governor in Council may, on the recommendation of the Minister, by order, provide that sections 32 and 33 apply in lands in a province that are not federal lands with respect to individuals of a listed wildlife species that is not an aquatic species or a species of birds that are migratory birds protected by the <i>Migratory Birds Convention Act, 1994</i> .
Obligation to make recommendation	(3) The Minister must recommend that the order be made if the Minister is of the opinion that the laws of the province do not protect the species.

Consultation

(4) Before recommending that the Governor in Council make an order under subsection (2), the Minister must consult

(a) the appropriate provincial minister; and

(b) if the species is found in an area in respect of which a wildlife management board is authorized by a land claims agreement to perform functions in respect of wildlife species, the wildlife management board.



Glossary COSEWIC

amphibian	Any cold-blooded vertebrate of the class Amphibia, typically living on land but breeding in water.
biodiversity	The variety of life (species and genetic stocks) in different ecosystems; "biodiversity" is short for biological diversity.
ecosystem	All organisms in a community, plus the associated environmental factors with which they interact.
environment	The components of the Earth, including: land, water and air; all organic and inorganic matter and living organisms; and the interacting natural systems of which these components are a part.
endangered	A species facing imminent extirpation or extinction.
extinct	A species that no longer exists.
extirpated	A species that no longer exists in the wild in Canada but occurs elsewhere (e.g., in captivity or in the wild in the United States).
genus	A category of biological classification comprised of a group of closely related or structurally similar species, or a single species exhibiting unusual features; appears as the first half of the scientific name of a species (e.g., in the scientific name <i>Grus americana</i> , <i>Grus</i> is the genus).
habitat	The particular type of local environment occupied by an individual or a population.
hibernation	A state of inactivity or dormancy assumed by certain animals during periods of unfavourable climatic conditions; while hibernating, an animal's metabolic processes are greatly slowed.
indeterminate	A species for which there is insufficient scientific information to support status designation.
lichen	Any of various small plants, which are formed by the symbiotic association of a fungus and an alga and occur as crusty patches or bushy growths on trees or rocks.
mammal	Any animal of the Mammalia, a large class of warm-blooded vertebrates having mammary glands.
migration	The periodical movement of an individual from one locality or climate to another for feeding, breeding or hibernating (e.g., some bat species).
mollusc	Any of various invertebrates having a soft unsegmented body, and often a shell secreted by a fold of skin (called the mantle). The group includes the gastropods (snails, slugs, etc.), bivalves (clams, mussels, etc.), and cephalopods (cuttlefish, octopuses, etc.).
moss	Any of a class of plants, typically growing in dense mats on trees, rocks, moist ground, etc.
organism	Any individual living creature, either unicellular (of one cell) or multicellular.
pesticide	A general term for a chemical agent used to kill pests (organisms unwanted by humans at a specific time and place); compare with insecticide, a chemical agent used to kill insects and invertebrate pests; and herbicide, a chemical agent used to kill weeds or herbage.
population	A group of individuals of one species found within a prescribed area and usually somewhat isolated from other groups of the same species.
range	The geographical limits of a species or group; a migratory species usually has both a breeding range and a wintering range.
recovery	Conservation actions undertaken to benefit an endangered or threatened species until a sustainable population level of fit individuals has been reached and threats to the population have been controlled.
reptile	Any of the cold-blooded vertebrates, of the class Reptilia, characterized by lungs, an outer covering of horny scales or plates, and young produced in eggs.

restoration	Conservation actions undertaken to increase the population size and distribution of an endangered or threatened species to what existed before human activities caused the population to decline.
special concern	A species of special concern because of characteristics that make it particularly sensitive to human activities or natural events.
species	A group of closely related organisms which are potentially capable of interbreeding, and which are reproductively isolated from other groups of organisms; the basic unit of biological classification. Each species is designated by an italicized, binomial Latin name that consists of the genus (capitalized) followed by a noun or adjective (uncapitalized) (e.g., in the scientific name <i>Grus americana</i> , <i>americana</i> is the species).
subspecies	A geographically limited subdivision of a species that is taxonomically different from other such subdivisions of the same species.
sustainable use	Human use of an organism or ecosystem at a rate within its capacity for renewal or regeneration.
taxonomy	The science of classification, esp. of living and extinct organisms
threatened	A species likely to become endangered if limiting factors are not reversed.
vertebrate	Any animal of a subphylum characterized by a bony skeleton and a well-developed brain: the group contains fishes, amphibians, reptiles, birds and mammals.
wetland	Land that is saturated with water long enough to promote wetland or aquatic processes; examples are bogs, fens, marshes, swamps, and shallow waters.
wildlife	Any wild organism, including: wild mammals, birds, reptiles, amphibians, fishes, invertebrates, plants, fungi, algae, and bacteria.